P8157 Homework 2

Jiaxin Zhang (jz4027)

Question 1

```
load("MACS-VL.RData")
dat <- macsVL %>%
  arrange(id, month) %>%
  mutate(lvl = log10(vload),   # log10 baseline viral load
         x   = lvl - 4)        # centered at 10,000 copies/mL
subj <- dat %>% group_by(id) %>%
  summarise(vload = first(vload), lvl = first(lvl), n_obs = n(),
            fu = max(month), cd4_base = cd4[which.min(month)])
```

(a) Descriptive summaries

```
c(subjects = nrow(subj), observations = nrow(dat))
```

```
##     subjects observations
##          225         1684
```

```
summary(subj$n_obs)   # measurements per subject
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.000   7.000   8.000   7.484   9.000  10.000
```

```
summary(subj$fu)      # follow-up (months since seroconversion)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.00   42.00   45.00   42.22   47.00   48.00
```

```
summary(subj$vload)   # baseline viral load (copies/mL)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     300    7928   24573   78348   91195 1026656
```

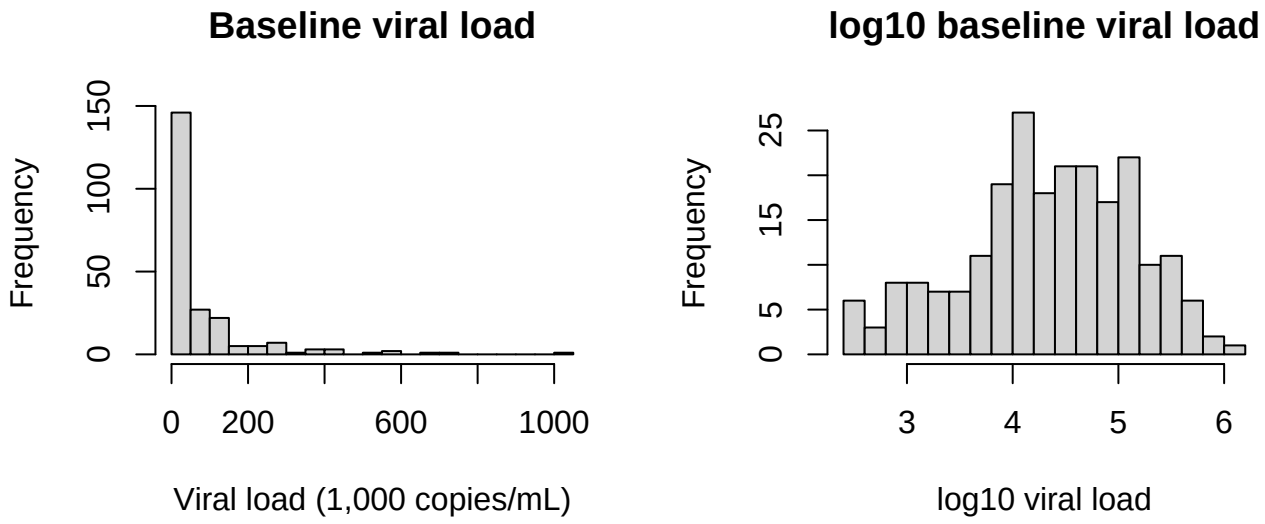
```
summary(subj$lvl)     # log10 baseline viral load
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.477   3.899   4.390   4.341   4.960   6.011
```

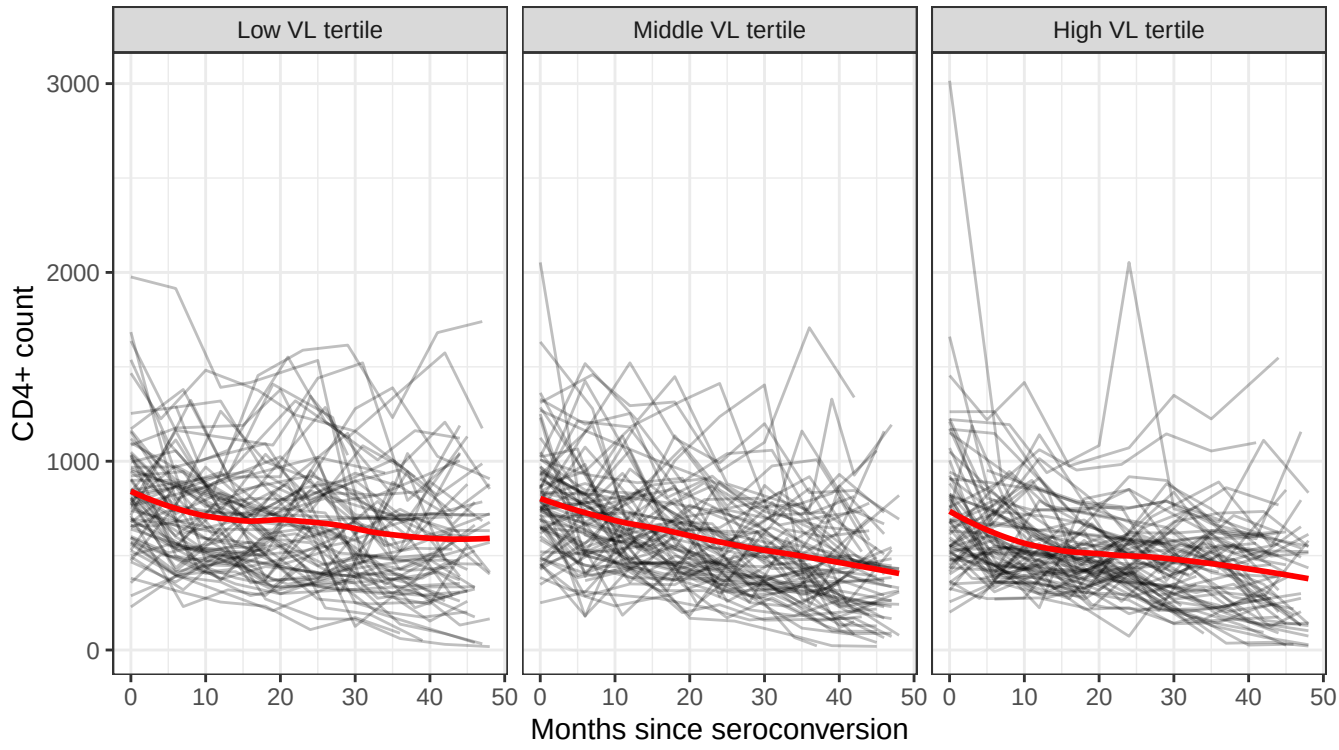
```r
summary(dat$cd4)      # CD4+ count (all visits)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    18.0   387.0   551.0   603.2   780.2  3015.0
```

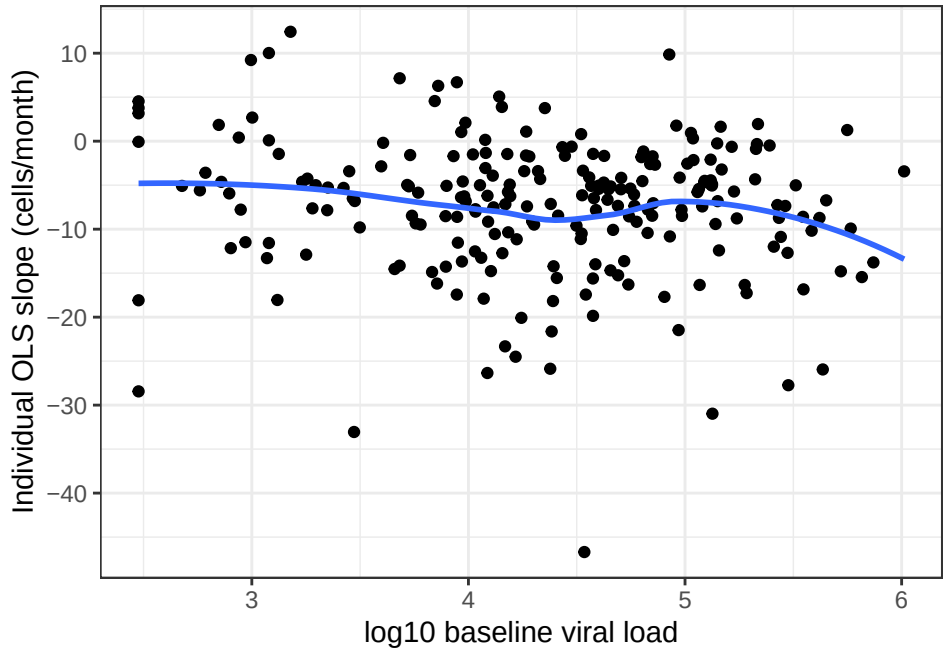
```r
par(mfrow = c(1, 2))
hist(subj$vload / 1000, breaks = 30, main = "Baseline viral load",
     xlab = "Viral load (1,000 copies/mL)")
hist(subj$lvl, breaks = 20, main = "log10 baseline viral load",
     xlab = "log10 viral load")
```




```r
dat$vl_tert <- cut(dat$lvl, quantile(subj$lvl, c(0, 1/3, 2/3, 1)),
                   include.lowest = TRUE,
                   labels = c("Low VL tertile", "Middle VL tertile", "High VL tertile"))
ggplot(dat, aes(month, cd4)) +
  geom_line(aes(group = id), alpha = 0.25) +
  geom_smooth(method = "loess", se = FALSE, colour = "red") +
  facet_wrap(~ vl_tert) +
  labs(x = "Months since seroconversion", y = "CD4+ count") +
  theme_bw()
```




```
# subject-specific OLS slopes vs baseline viral load
subj$slope <- sapply(split(dat, dat$id), function(d) unname(coef(lm(cd4 ~ month, data = d))[2]))
ggplot(subj, aes(lvl, slope)) + geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(x = "log10 baseline viral load", y = "Individual OLS slope (cells/month)") +
  theme_bw()
```



```r
cor(subj$lvl, subj$slope)
```

```
## [1] -0.1269633
```

```r
# median individual slope by viral load tertile
tapply(subj$slope, cut(subj$lvl, quantile(subj$lvl, c(0, 1/3, 2/3, 1)),
                       include.lowest = TRUE), median)
```

```
## [2.48,4.07] (4.07,4.74] (4.74,6.01]
##   -5.586868   -7.135784   -6.725965
```

Summary. $K = 225$ men contribute 1,684 CD4+ measurements (median 8 per subject) at irregular times over up to 48 months. Baseline viral load is highly right-skewed (median 24,573 copies/mL, range 300–1,026,656), so we use $\log_{10}$ viral load, which is roughly symmetric. CD4+ declines over time in all viral load tertiles; higher viral load is associated with lower CD4+ at seroconversion and slightly steeper individual slopes (median -5.6, -7.1, -6.7 cells/month by tertile; correlation with $\log_{10}$ VL $= -0.13$).

(b) Exploratory analysis of the covariance structure

Residuals from an OLS fit of the mean model are grouped into 6-month windows (visit times are irregular); we also compute the sample variogram.

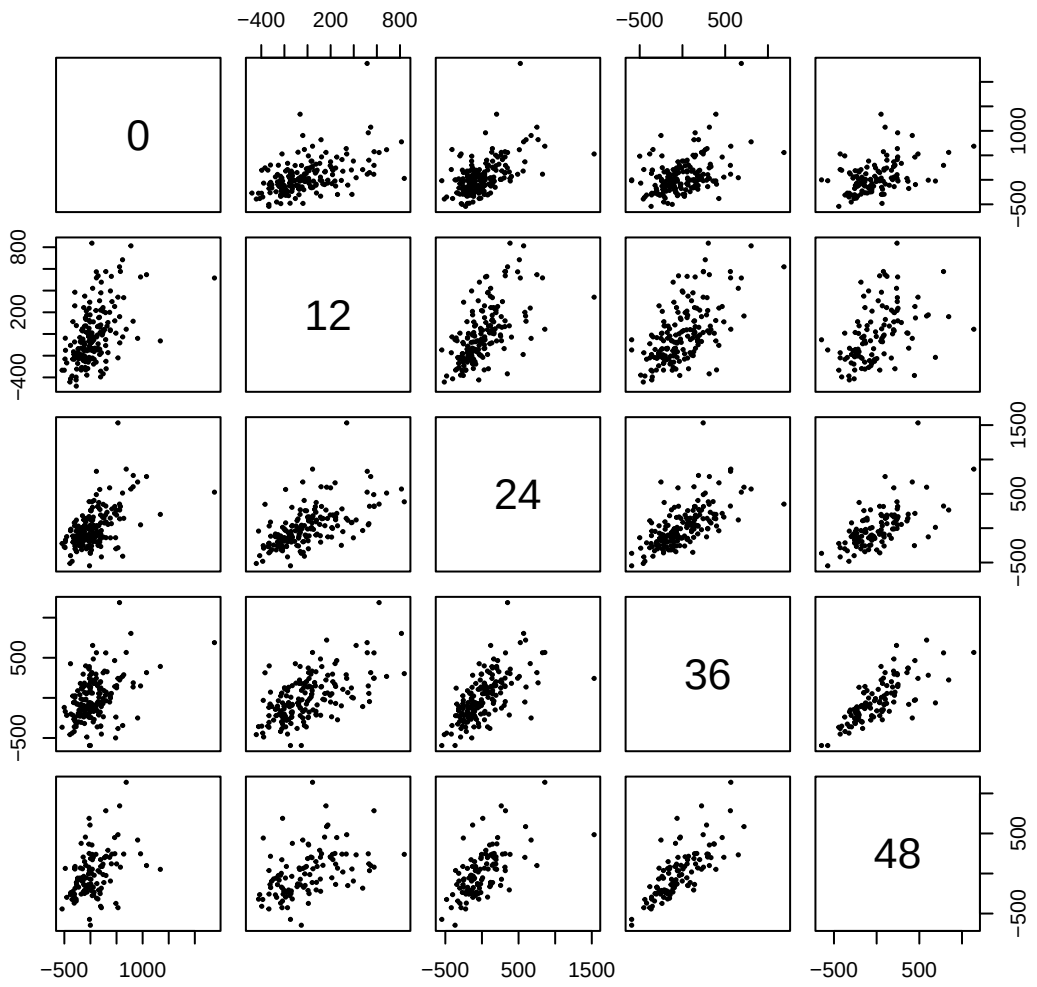
```r
dat$res <- resid(lm(cd4 ~ month * x, data = dat))
dat$tbin <- 6 * round(dat$month / 6)
wide <- dat %>% group_by(id, tbin) %>% summarise(res = mean(res), .groups = "drop") %>%
  pivot_wider(names_from = tbin, values_from = res) %>% select(-id)
wide <- wide[, order(as.numeric(names(wide)))]
round(sapply(wide, sd, na.rm = TRUE))                    # SD by time window
```

```
##   0   6  12  18  24  30  36  42  48
## 343 281 251 261 286 279 275 301 294
```

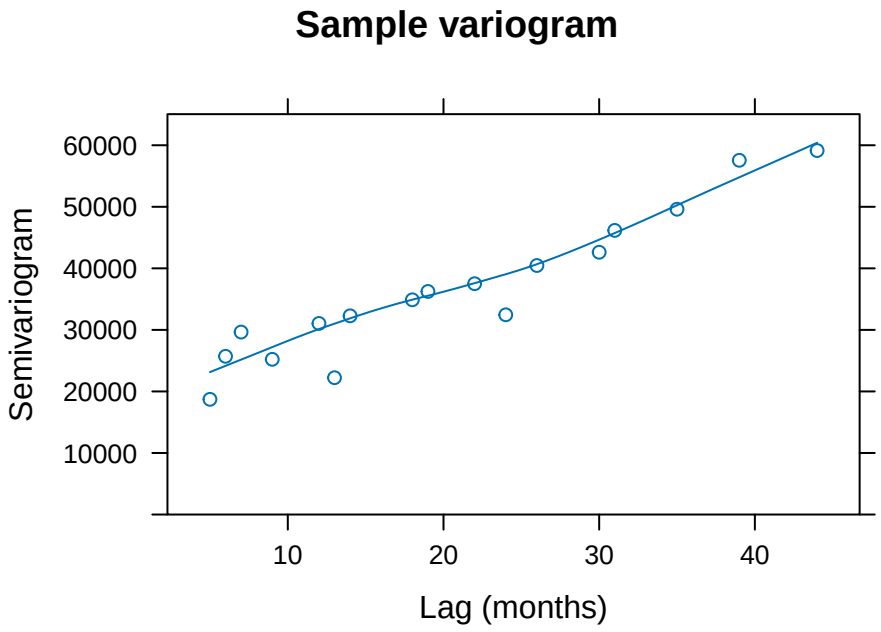
```r
round(cor(wide, use = "pairwise.complete.obs"), 2)       # correlation matrix
```

```
##       0    6   12   18   24   30   36   42   48
## 0  1.00 0.56 0.50 0.46 0.53 0.49 0.41 0.42 0.39
## 6  0.56 1.00 0.66 0.69 0.61 0.51 0.54 0.50 0.50
## 12 0.50 0.66 1.00 0.72 0.61 0.57 0.59 0.52 0.44
## 18 0.46 0.69 0.72 1.00 0.74 0.71 0.67 0.62 0.60
## 24 0.53 0.61 0.61 0.74 1.00 0.69 0.64 0.66 0.62
## 30 0.49 0.51 0.57 0.71 0.69 1.00 0.72 0.77 0.72
## 36 0.41 0.54 0.59 0.67 0.64 0.72 1.00 0.79 0.75
## 42 0.42 0.50 0.52 0.62 0.66 0.77 0.79 1.00 0.84
## 48 0.39 0.50 0.44 0.60 0.62 0.72 0.75 0.84 1.00
```

```r
pairs(wide[, c("0", "12", "24", "36", "48")], pch = 20, cex = 0.5,
      main = "Residuals at months 0, 12, 24, 36, 48")
```




```
fit0 <- gls(cd4 ~ month * x, data = dat)
vg <- Variogram(fit0, form = ~ month | id, resType = "response")
plot(vg, ylim = c(0, max(vg$variog) * 1.1),
     xlab = "Lag (months)", main = "Sample variogram")
```




```
# process variance (total variance of residuals) for reference
var(dat$res)
```

```
## [1] 82552.92
```

Interpretation. The residual SD is roughly constant over time (250–340). Correlations are high (0.39–0.84) and decrease with lag but do not approach zero; the variogram likewise rises with lag but stays well below the total variance (≈ 82,500).

Hence **compound symmetry** is a reasonable approximation, while an **exponential (AR(1)-type) correlation plus a random intercept**, or a **random intercept and slope** model, would capture the decay with lag better. Pure AR(1) is not plausible since correlations do not decay to zero.

(c) Linear model in log viral load, compound symmetry, ML vs REML

We use $x_k = \log_{10}(\text{baseline VL}) - 4$ (centered at 10,000 copies/mL).

```
fit_ml   <- gls(cd4 ~ month * x, data = dat, method = "ML",
                correlation = corCompSymm(form = ~ 1 | id))
fit_reml <- gls(cd4 ~ month * x, data = dat, method = "REML",
                correlation = corCompSymm(form = ~ 1 | id))

tab_fit <- function(fit) {
  ct <- summary(fit)$tTable
  ci <- confint(fit)    # Wald 95% CI
  data.frame(Est = ct[, 1], SE = ct[, 2], Lower = ci[, 1], Upper = ci[, 2], p = ct[, 4])
}
tab_c <- cbind(tab_fit(fit_ml), tab_fit(fit_reml))
rownames(tab_c) <- c("Intercept (beta0)", "Month (beta1)",
                     "log10 VL (beta2)", "Month x log10 VL (beta3)")
```

```r
vc <- function(fit) c(sigma = fit$sigma,
                      rho = coef(fit$modelStruct$corStruct, unconstrained = FALSE),
                      logLik = as.numeric(logLik(fit)))
kable(tab_c, digits = c(2, 2, 2, 2, 4, 2, 2, 2, 2, 4),
      format = "latex", booktabs = TRUE,
      col.names = rep(c("Est.", "SE", "2.5%", "97.5%", "p"), 2),
      caption = "GLS estimates with compound symmetric correlation: ML (left) and REML (right).")
```

Table 1: GLS estimates with compound symmetric correlation: ML (left) and REML (right).

	Est.	SE	2.5%	97.5%	p	Est.	SE	2.5%	97.5%	p
Intercept (beta0)	779.27	17.87	744.23	814.30	0.0000	779.27	17.92	744.14	814.40	0.0000
Month (beta1)	-6.59	0.34	-7.26	-5.92	0.0000	-6.59	0.34	-7.26	-5.92	0.0000
log10 VL (beta2)	-82.21	20.71	-122.79	-41.63	0.0001	-82.21	20.76	-122.89	-41.52	0.0001
Month x log10 VL (beta3)	-0.88	0.40	-1.66	-0.09	0.0286	-0.88	0.40	-1.66	-0.09	0.0286

```r
kable(rbind(ML = vc(fit_ml), REML = vc(fit_reml)), digits = 3, format = "latex", booktabs = TRUE,
      col.names = c("sigma", "rho", "log-lik"),
      caption = "Variance components.")
```

Table 2: Variance components.

	sigma	rho	log-lik
ML	283.383	0.567	-11458.28
REML	284.249	0.569	-11451.02

Results. ML and REML estimates are practically identical; REML gives a slightly larger $\hat{\sigma}$ (284.2 vs 283.4), as expected; the SEs are nearly identical. Within-subject correlation $\hat{\rho} \approx 0.57$.

Interpretation (REML estimates, 95% CI).

- $\hat{\beta}_0 = 779.3$ (744.1, 814.4): estimated mean CD4+ count at seroconversion ($t = 0$) for men with a baseline viral load of 10,000 copies/mL ($\log_{10}$VL $= 4$).
- $\hat{\beta}_1 = -6.59$ (-7.26, -5.92): among men with a baseline viral load of 10,000 copies/mL, mean CD4+ count declines by an estimated 6.6 cells per month (about 79 cells per year).
- $\hat{\beta}_2 = -82.2$ (-122.9, -41.5): at seroconversion, mean CD4+ count is 82.2 cells lower for each 1-unit higher $\log_{10}$ baseline viral load (i.e. each 10-fold higher viral load).
- $\hat{\beta}_3 = -0.88$ (-1.66, -0.09): the rate of decline in mean CD4+ count is 0.88 cells/month steeper for each 10-fold higher baseline viral load.

Since $p = 0.029$ for β_3 (both ML and REML), there is a significant association between higher baseline viral load and a faster rate of CD4+ decline.

(d) Categorized baseline viral load (REML, compound symmetry)

Baseline viral load is categorized by quartiles ($J = 4$; Q1 = reference).

```r
q <- quantile(subj$vload, c(0, .25, .5, .75, 1))
q
```

```
##      0%     25%     50%     75%    100%
##     300    7928   24573   91195 1026656
```

```r
dat$vlcat <- cut(dat$vload, q, include.lowest = TRUE,
                 labels = c("Q1", "Q2", "Q3", "Q4"))
table(subj = cut(subj$vload, q, include.lowest = TRUE, labels = c("Q1","Q2","Q3","Q4")))
```

```
## subj
## Q1 Q2 Q3 Q4
## 57 56 56 56
```

```r
fit_d <- gls(cd4 ~ month * vlcat, data = dat, method = "REML",
             correlation = corCompSymm(form = ~ 1 | id))
tab_d <- tab_fit(fit_d)
rownames(tab_d) <- c("Intercept (beta0)", "Month (beta1)",
                     "Q2 (beta22)", "Q3 (beta23)", "Q4 (beta24)",
                     "Month x Q2 (beta32)", "Month x Q3 (beta33)", "Month x Q4 (beta34)")
kable(tab_d, digits = c(2, 2, 2, 2, 4), format = "latex", booktabs = TRUE,
      col.names = c("Est.", "SE", "2.5%", "97.5%", "p"),
      caption = paste0("REML GLS estimates, compound symmetry (sigma = ",
                       round(fit_d$sigma, 1), ", rho = ",
                       round(coef(fit_d$modelStruct$corStruct, unconstrained = FALSE), 3), ")."))
```

Table 3: REML GLS estimates, compound symmetry (sigma = 285, rho = 0.572).

	Est.	SE	2.5%	97.5%	p
Intercept (beta0)	859.87	32.91	795.38	924.37	0.0000
Month (beta1)	-5.64	0.62	-6.86	-4.43	0.0000
Q2 (beta22)	-112.75	46.61	-204.10	-21.39	0.0157
Q3 (beta23)	-146.08	46.71	-237.63	-54.53	0.0018
Q4 (beta24)	-177.75	46.81	-269.50	-86.00	0.0002
Month x Q2 (beta32)	-1.93	0.88	-3.66	-0.19	0.0295
Month x Q3 (beta33)	-1.81	0.87	-3.51	-0.10	0.0384
Month x Q4 (beta34)	-1.26	0.91	-3.03	0.51	0.1641

```r
# Slope (cells/month) within each viral load category
L <- rbind(Q1 = c(0, 1, 0, 0, 0, 0, 0, 0),
           Q2 = c(0, 1, 0, 0, 0, 1, 0, 0),
           Q3 = c(0, 1, 0, 0, 0, 0, 1, 0),
           Q4 = c(0, 1, 0, 0, 0, 0, 0, 1))
est <- drop(L %*% coef(fit_d)); se <- sqrt(diag(L %*% vcov(fit_d) %*% t(L)))
round(cbind(slope = est, SE = se, lower = est - 1.96 * se, upper = est + 1.96 * se), 2)
```

```
##    slope   SE lower upper
## Q1 -5.64 0.62 -6.86 -4.43
```

```
## Q2 -7.57 0.63 -8.81 -6.33
## Q3 -7.45 0.61 -8.65 -6.25
## Q4 -6.90 0.66 -8.20 -5.61
```

```
# Joint Wald test of H0: beta_32 = beta_33 = beta_34 = 0 (slopes equal across categories)
anova(fit_d, Terms = "month:vlcat")
```

```
## Denom. DF: 1676
##  F-test for: month:vlcat
##   numDF  F-value p-value
## 1     3 2.016845  0.1096
```

Interpretation (95% CI). Quartiles: Q1 300–7,928; Q2 7,928–24,573; Q3 24,573–91,195; Q4 91,195–1,026,656 copies/mL.

- $\hat{\beta}_0 = 859.9$ (795.4, 924.4): estimated mean CD4+ count at seroconversion in the lowest viral load quartile (Q1).
- $\hat{\beta}_1 = -5.64$ (-6.86, -4.43): estimated rate of change in mean CD4+ count in Q1, a decline of 5.6 cells per month.
- $\hat{\beta}_{2j}$: difference in mean CD4+ at seroconversion between quartile j and Q1: -112.7 (Q2), -146.1 (Q3), -177.8 (Q4) cells.
- $\hat{\beta}_{3j}$: difference in the monthly rate of change between quartile j and Q1: -1.93 (Q2), -1.81 (Q3), -1.26 (Q4) cells/month, i.e. slopes of -5.64, -7.57, -7.45, -6.90 in Q1–Q4.

The joint Wald test of $H_0 : \beta_{32} = \beta_{33} = \beta_{34} = 0$ gives $F_{3,1676} = 2.02$, $p = 0.11$, so there is no significant association between categorized baseline viral load and the rate of CD4+ decline in this model (although Q2 and Q3 individually differ from Q1).

(e) Varying coefficient model with natural cubic splines

$\gamma_0(x)$ and $\gamma_1(x)$ are natural cubic splines in $\log_{10}$ VL with knots at the tertiles, fit by REML with compound symmetry:

$$E[Y_{ki}] = \gamma_0(x_k) + \gamma_1(x_k)t_{ki}, \qquad \gamma_j(x) = \alpha_{j0} + \sum_{l=1}^{3} \alpha_{jl} B_l(x), \; j = 0, 1.$$

```
kn  <- quantile(subj$lvl, c(1/3, 2/3))
kn
```

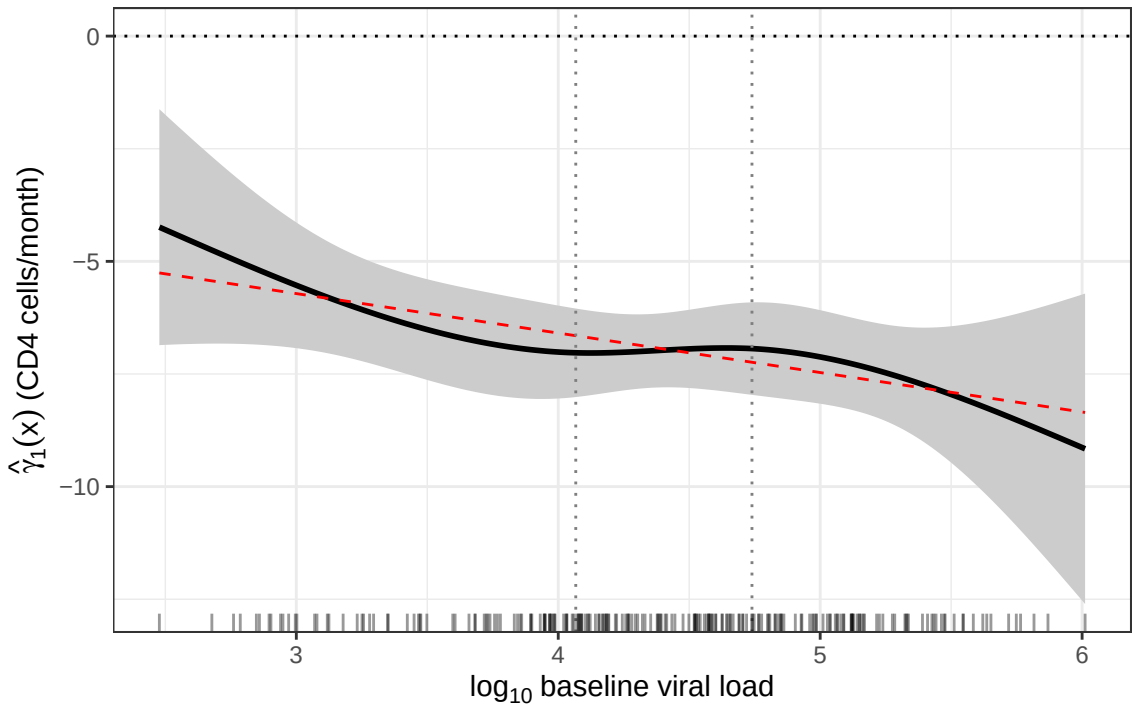
```
## 33.33333% 66.66667%
##  4.066864  4.739628
```

```
nsb <- ns(subj$lvl, knots = kn)
B   <- predict(nsb, dat$lvl); colnames(B) <- paste0("B", 1:3)
dat <- cbind(dat, B)
fit_e <- gls(cd4 ~ (B1 + B2 + B3) * month, data = dat, method = "REML",
             correlation = corCompSymm(form = ~ 1 | id))

# gamma_1(x) on a grid with pointwise 95% CI
grid <- seq(min(subj$lvl), max(subj$lvl), length.out = 200)
idx  <- grep("month", names(coef(fit_e)))          # month, B1:month, B2:month, B3:month
```

```
Lg   <- cbind(1, predict(nsb, grid))
g1   <- drop(Lg %*% coef(fit_e)[idx])
g1se <- sqrt(rowSums((Lg %*% vcov(fit_e)[idx, idx]) * Lg))
pe   <- data.frame(lvl = grid, g1 = g1, lo = g1 - 1.96 * g1se, hi = g1 + 1.96 * g1se,
                   lin = coef(fit_reml)[2] + coef(fit_reml)[4] * (grid - 4))
```

```
ggplot(pe, aes(lvl)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), fill = "grey80") +
  geom_line(aes(y = g1), linewidth = 1) +
  geom_line(aes(y = lin), linetype = "dashed", colour = "red") +
  geom_hline(yintercept = 0, linetype = "dotted") +
  geom_vline(xintercept = kn, linetype = "dotted", colour = "grey50") +
  geom_rug(data = subj, aes(x = lvl), inherit.aes = FALSE, alpha = 0.4) +
  labs(x = expression(log[10]~"baseline viral load"),
       y = expression(hat(gamma)[1](x)~"(CD4 cells/month)"),
       caption = paste("Solid: spline estimate with pointwise 95% CI;",
                       "dashed: linear model from (c) (REML); dotted: knots")) +
  theme_bw()
```



Solid: spline estimate with pointwise 95% CI; dashed: linear model from (c) (REML); dotted: knots

```
# gamma_1 at selected viral loads
xv <- log10(c(1000, 5000, 10000, 50000, 100000, 500000))
Lv <- cbind(1, predict(nsb, xv))
v  <- drop(Lv %*% coef(fit_e)[idx]); vse <- sqrt(rowSums((Lv %*% vcov(fit_e)[idx, idx]) * Lv))
kable(data.frame(VL = format(10^xv, big.mark = ",", scientific = FALSE),
                 spline = v, lower = v - 1.96 * vse, upper = v + 1.96 * vse,
                 linear_c = coef(fit_reml)[2] + coef(fit_reml)[4] * (xv - 4)),
      digits = 2, format = "latex", booktabs = TRUE, row.names = FALSE,
```

```
    col.names = c("Baseline VL", "gamma1 (spline)", "2.5%", "97.5%", "gamma1 (model c)"),
    caption = "Estimated rate of change in CD4+ (cells/month) at selected baseline viral loads.")
```

Table 4: Estimated rate of change in CD4+ (cells/month) at selected baseline viral loads.

Baseline VL	gamma1 (spline)	2.5%	97.5%	gamma1 (model c)
1,000	-5.53	-6.93	-4.14	-5.72
5,000	-6.78	-7.92	-5.65	-6.33
10,000	-7.01	-8.04	-5.98	-6.59
50,000	-6.93	-7.93	-5.92	-7.20
100,000	-7.12	-8.16	-6.08	-7.47
500,000	-8.40	-10.56	-6.24	-8.08

```
# Model (c) is nested in the spline model: likelihood ratio test using ML fits
fit_e_ml <- update(fit_e, method = "ML")
anova(fit_ml, fit_e_ml)
```

```
##          Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## fit_ml       1  6 22928.55 22961.12 -11458.28
## fit_e_ml     2 10 22931.12 22985.42 -11455.56 1 vs 2 5.426297  0.2463
```

Interpretation. $\hat{\gamma}_1(x)$ is the estimated rate of change in mean CD4+ (cells/month) at baseline $\log_{10}$ VL x; it is negative throughout. The decline is 5.5 cells/month (95% CI 4.1, 6.9) at 1,000 copies/mL, 7.0 (6.0, 8.0) at 10,000, roughly flat up to 100,000 (7.1), and 8.4 (6.2, 10.6) at 500,000 copies/mL.

Adequacy of model (c). Apart from a mild plateau between 10^4 and 10^5 copies/mL, the linear fit from (c) lies within the 95% band everywhere (LRT vs. spline model $p = 0.25$), suggesting model (c) is adequate.

Question 2

Notation. Take $m = n$ and $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$. Let

$$\bar{Y}_{k\cdot} = \frac{1}{n}\sum_i Y_{ki}, \quad \bar{Y}_{\cdot\cdot} = \frac{1}{K}\sum_k \bar{Y}_{k\cdot}, \quad \text{SSE} = \sum_k\sum_i (Y_{ki} - \bar{Y}_{k\cdot})^2, \quad \text{SSA} = n\sum_k (\bar{Y}_{k\cdot} - \bar{Y}_{\cdot\cdot})^2,$$

so $\text{MSE} = \text{SSE}/[K(n-1)]$ and $\text{MSA} = \text{SSA}/(K-1)$, and let $\lambda = \sigma^2 + n\tau^2$.

(a) Likelihood and log-likelihood

For unit k, $\mathbf{Y}_k = (Y_{k1}, \ldots, Y_{kn})^\top = \mu\mathbf{1}_n + \gamma_k\mathbf{1}_n + \boldsymbol{\epsilon}_k$, so

$$\mathbf{Y}_k \sim N_n(\mu\mathbf{1}_n,\ \mathbf{V}), \qquad \mathbf{V} = \sigma^2\mathbf{I}_n + \tau^2\mathbf{J}_n,$$

independently over k. Using the results given with $a = \sigma^2$, $b = \tau^2$:

$$\mathbf{V}^{-1} = \frac{1}{\sigma^2}\left(\mathbf{I}_n - \frac{\tau^2}{\lambda}\mathbf{J}_n\right), \qquad |\mathbf{V}| = (\sigma^2)^{n-1}\lambda.$$

For the quadratic form, write $\mathbf{r}_k = \mathbf{Y}_k - \mu\mathbf{1}_n$, so that $\mathbf{1}_n^\top\mathbf{r}_k = n(\bar{Y}_{k\cdot} - \mu)$ and $\mathbf{r}_k^\top\mathbf{r}_k = \sum_i (Y_{ki} - \bar{Y}_{k\cdot})^2 + n(\bar{Y}_{k\cdot} - \mu)^2$. Then

$$\begin{aligned}
\mathbf{r}_k^\top\mathbf{V}^{-1}\mathbf{r}_k &= \frac{1}{\sigma^2}\left[\sum_i (Y_{ki} - \bar{Y}_{k\cdot})^2 + n(\bar{Y}_{k\cdot} - \mu)^2 - \frac{\tau^2}{\lambda}n^2(\bar{Y}_{k\cdot} - \mu)^2\right] \\
&= \frac{1}{\sigma^2}\sum_i (Y_{ki} - \bar{Y}_{k\cdot})^2 + \frac{n(\bar{Y}_{k\cdot} - \mu)^2}{\sigma^2}\cdot\frac{\lambda - n\tau^2}{\lambda} = \frac{1}{\sigma^2}\sum_i (Y_{ki} - \bar{Y}_{k\cdot})^2 + \frac{n(\bar{Y}_{k\cdot} - \mu)^2}{\lambda}.
\end{aligned}$$

Also $\sum_k n(\bar{Y}_{k\cdot} - \mu)^2 = \text{SSA} + nK(\bar{Y}_{\cdot\cdot} - \mu)^2$. Hence the likelihood is

$$\begin{aligned}
L(\mu, \sigma^2, \tau^2) &= \prod_{k=1}^K (2\pi)^{-n/2}|\mathbf{V}|^{-1/2}\exp\left\{-\tfrac{1}{2}\mathbf{r}_k^\top\mathbf{V}^{-1}\mathbf{r}_k\right\} \\
&= (2\pi)^{-Kn/2}(\sigma^2)^{-K(n-1)/2}(\sigma^2 + n\tau^2)^{-K/2}\exp\left\{-\frac{\text{SSE}}{2\sigma^2} - \frac{\text{SSA} + nK(\bar{Y}_{\cdot\cdot} - \mu)^2}{2(\sigma^2 + n\tau^2)}\right\},
\end{aligned}$$

and the log-likelihood is

$$\ell(\mu, \sigma^2, \tau^2) = -\frac{Kn}{2}\log(2\pi) - \frac{K(n-1)}{2}\log\sigma^2 - \frac{K}{2}\log(\sigma^2 + n\tau^2) - \frac{\text{SSE}}{2\sigma^2} - \frac{\text{SSA} + nK(\bar{Y}_{\cdot\cdot} - \mu)^2}{2(\sigma^2 + n\tau^2)}.$$

(b) Maximum likelihood estimators

The map $(\sigma^2, \tau^2) \mapsto (\sigma^2, \lambda)$ is one-to-one, so by invariance we may maximize over (μ, σ^2, λ):

$$\ell = c - \frac{K(n-1)}{2}\log\sigma^2 - \frac{K}{2}\log\lambda - \frac{\text{SSE}}{2\sigma^2} - \frac{\text{SSA} + nK(\bar{Y}_{\cdot\cdot} - \mu)^2}{2\lambda}.$$

- μ: $\dfrac{\partial\ell}{\partial\mu} = \dfrac{nK(\bar{Y}_{\cdot\cdot} - \mu)}{\lambda} = 0 \ \Rightarrow\ \hat{\mu} = \bar{Y}_{\cdot\cdot}$ (for any σ^2, λ; ℓ is concave in μ).
- σ^2: $\dfrac{\partial\ell}{\partial\sigma^2} = -\dfrac{K(n-1)}{2\sigma^2} + \dfrac{\text{SSE}}{2\sigma^4} = 0 \ \Rightarrow\ \hat{\sigma}^2 = \dfrac{\text{SSE}}{K(n-1)} = \text{MSE}.$

- λ (at $\mu = \hat{\mu}$): $\dfrac{\partial \ell}{\partial \lambda} = -\dfrac{K}{2\lambda} + \dfrac{\mathrm{SSA}}{2\lambda^2} = 0 \;\Rightarrow\; \hat{\lambda} = \dfrac{\mathrm{SSA}}{K} = \dfrac{K-1}{K}\mathrm{MSA} = \left(1 - \tfrac{1}{K}\right)\mathrm{MSA}.$

These are maxima since $-\frac{a}{2}\log v - \frac{b}{2v}$ is maximized at $v = b/a$. Therefore

$$\hat{\mu} = \bar{Y}_{\cdot\cdot}, \qquad \hat{\sigma}^2 = \mathrm{MSE}, \qquad \hat{\tau}^2 = \frac{\hat{\lambda} - \hat{\sigma}^2}{n} = \frac{(1 - 1/K)\,\mathrm{MSA} - \mathrm{MSE}}{n}.$$

(This is the stated result with $m = n$.)

(c) Variance of $\hat{\mu}$

The $\bar{Y}_{k\cdot} = \mu + \gamma_k + \bar{\epsilon}_{k\cdot}$ are independent with $\mathrm{Var}(\bar{Y}_{k\cdot}) = \tau^2 + \sigma^2/n = \lambda/n$, so

$$\mathrm{Var}[\hat{\mu}] = \mathrm{Var}[\bar{Y}_{\cdot\cdot}] = \frac{1}{K}\left(\tau^2 + \frac{\sigma^2}{n}\right) = \frac{\sigma^2 + n\tau^2}{nK}.$$

Plugging in the MLEs,

$$\widehat{\mathrm{Var}}[\hat{\mu}] = \frac{\hat{\sigma}^2 + n\hat{\tau}^2}{nK} = \frac{\hat{\lambda}}{nK} = \frac{(K-1)\,\mathrm{MSA}}{nK^2}.$$

(d) REML estimators

Integrate μ out of the likelihood in (a). Only the last term depends on μ, and

$$\int_{-\infty}^{\infty} \exp\left\{-\frac{nK(\bar{Y}_{\cdot\cdot} - \mu)^2}{2\lambda}\right\} d\mu = \left(\frac{2\pi\lambda}{nK}\right)^{1/2}.$$

Hence the restricted likelihood is

$$L_R(\sigma^2, \tau^2) = \int L\, d\mu = (2\pi)^{-(Kn-1)/2}(nK)^{-1/2}(\sigma^2)^{-K(n-1)/2}\lambda^{-(K-1)/2} \exp\left\{-\frac{\mathrm{SSE}}{2\sigma^2} - \frac{\mathrm{SSA}}{2\lambda}\right\},$$

$$\ell_R = c - \frac{K(n-1)}{2}\log\sigma^2 - \frac{K-1}{2}\log\lambda - \frac{\mathrm{SSE}}{2\sigma^2} - \frac{\mathrm{SSA}}{2\lambda}.$$

Setting derivatives to zero:

$$\frac{\partial \ell_R}{\partial \sigma^2} = -\frac{K(n-1)}{2\sigma^2} + \frac{\mathrm{SSE}}{2\sigma^4} = 0 \Rightarrow \tilde{\sigma}^2 = \mathrm{MSE}, \qquad \frac{\partial \ell_R}{\partial \lambda} = -\frac{K-1}{2\lambda} + \frac{\mathrm{SSA}}{2\lambda^2} = 0 \Rightarrow \tilde{\lambda} = \frac{\mathrm{SSA}}{K-1} = \mathrm{MSA}.$$

Therefore the REML estimators are

$$\tilde{\sigma}^2 = \mathrm{MSE}, \qquad \tilde{\tau}^2 = \frac{\mathrm{MSA} - \mathrm{MSE}}{n}$$

(These are the unbiased ANOVA estimators.)

(e) F-test of $H : \tau^2 = 0$

We are given that

$$\frac{\mathrm{MSA}/(\sigma^2 + n\tau^2)}{\mathrm{MSE}/\sigma^2} = \frac{\mathrm{MSA}}{\mathrm{MSE}} \cdot \frac{\sigma^2}{\sigma^2 + n\tau^2} \sim F_{K-1,\,K(n-1)}.$$

Under $H : \tau^2 = 0$ we have $\sigma^2 + n\tau^2 = \sigma^2$, so the unknown variances cancel and

$$F^\star = \frac{\mathrm{MSA}}{\mathrm{MSE}} \sim F_{K-1,\,K(n-1)} \quad \text{under } H.$$

So $F^\star$ has a known null distribution and can be compared with $F_{K-1,K(n-1)}$. Under $\tau^2 > 0$, $F^\star$ is inflated by the factor $1 + n\tau^2/\sigma^2$, so we reject H for large values: $F^\star > F_{K-1,K(n-1);1-\alpha}$.